

DR.I **STUDY DESIGN** SHA

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Planning ...(most critical elements)

⇒**Strategic planning**

Provides a framework for more detailed planning and day to day decisions Ensure right thing now & in the future determine mission & **long term** vision.

⇒**Operational planning**

Prepared perform the **immediate tasks**

⇒**Business Planning**

To determine **the feasibility**

⇒**Resources planning**

The resources necessary **to achieve the goals** and strategy of the organization.

⇒**Organizational planning**

Challenging of the **futures**

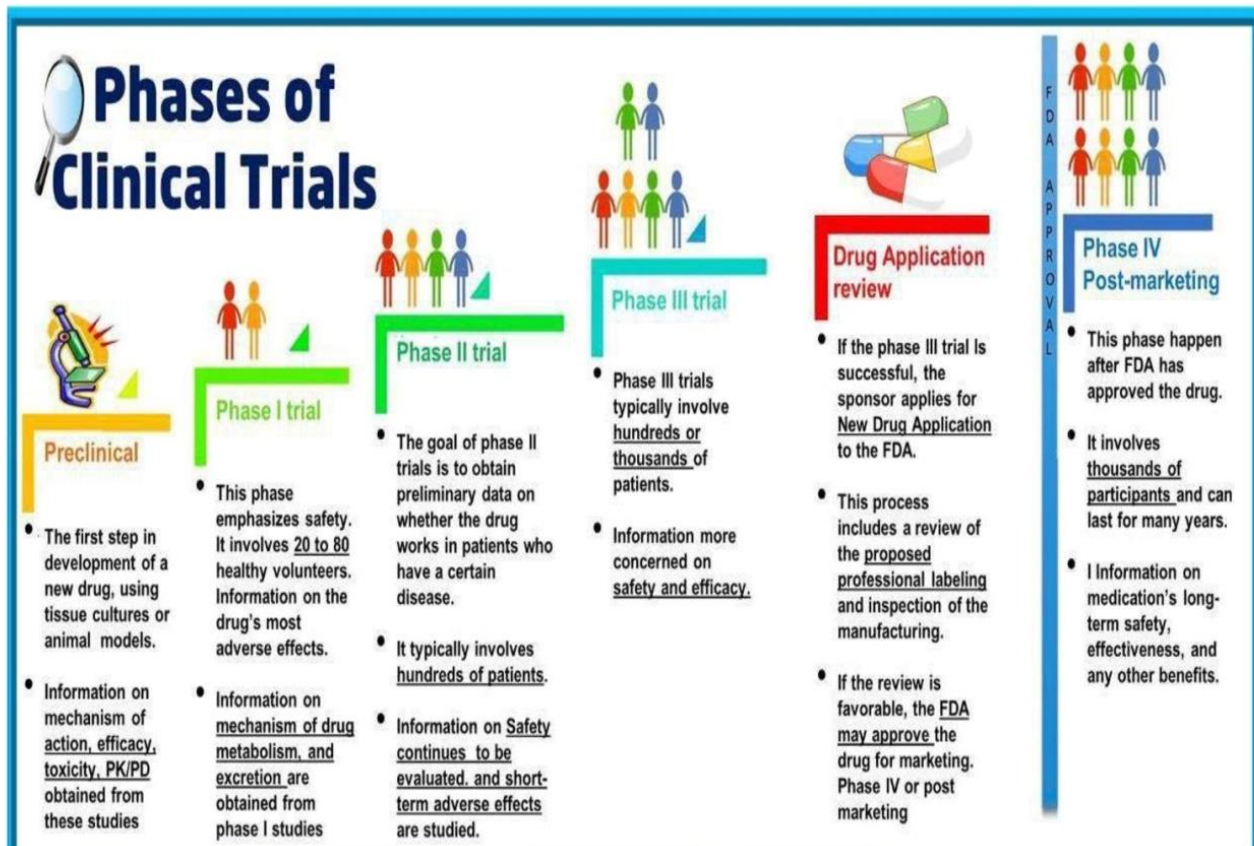
⇒**Contingency planning**

To provide a **fallback** option

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→Phase I for safety so must be on healthy volunteers.

→Phase II efficacy & S/E on pt

→Phase III RCT & FDA approval

→Phase IV Pharmacovigilance (post marketing)

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KEY WORDS

Incidence

The number of **NEW** cases in a population occurring in a given time period

Prevalence

The number of people in a population **with a disease** within a given time period

Mortality

The number of people **who have died** of a certain disease in a given time period

Endemic

An infectious disease that is **always present** in a population

Epidemic

When a disease **suddenly spreads** rapidly to affect many people (eg: influenza)

Pandemic

When a disease **spreads over a very large area** (eg: a continent or the whole world; we are having pandemics of AIDS and TB at present)

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$$\text{prev**A**lence} = \frac{\text{ALL cases}}{\text{Population @ risk}}$$

$$\text{i**N**cidence} = \frac{\text{New cases}}{\text{Population @ risk}}$$

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1. Incidence occur when the new cases is diagnosed

Each new case diagnosed



incidence

Prevalance

Disease is cured or Pt dies



prevalance—

Incidence not affected



Epidemiologist's Bathtub

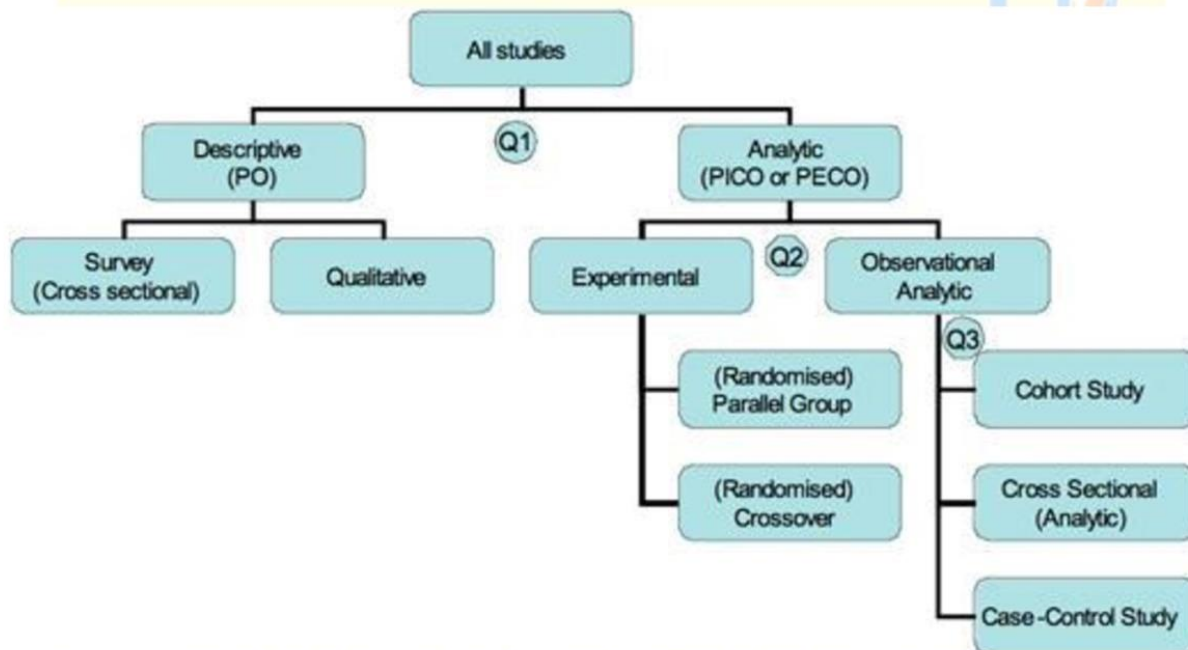


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❖ Study Design

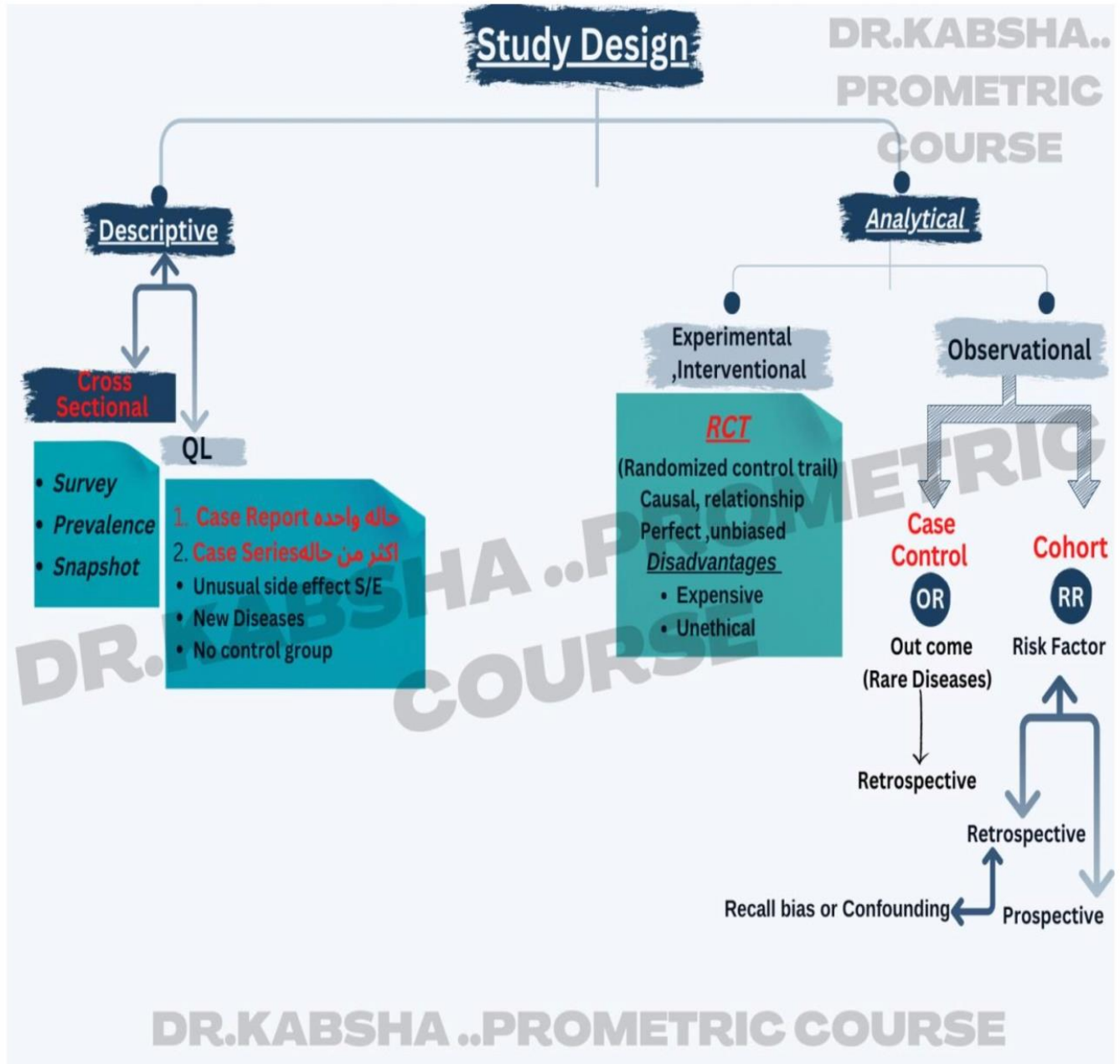
Definition

✓ Study design is an organized process used to evaluate a hypothesis to test whether it's accepted or rejected or not.



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➤ The used measure in **Case-Control** study is **Odds ratio (OR)**

OR = chance of event happening / chance of non event happening .

➤ **Interpretation** of the Odds Ratio :

✓ **OR = 1** → Exposure **NOT** related to disease(outcome)

✓ **OR >1** → Exposure **POSITIVELY** related to disease(outcome)

✓ **OR <1** → Exposure **NEGATIVELY** related to disease(outcome)

➤ The used measure in **Cohort** study is **Relative Risk (RR)**.

RR =incidence of outcome in exposed /incidence of outcome in non exposed

Interpretation of the Relative Risk :

✓ **RR = 1** → Exposure **NOT** related to disease(outcome)

✓ **RR >1** → Exposure **POSITIVELY** related to disease(outcome)

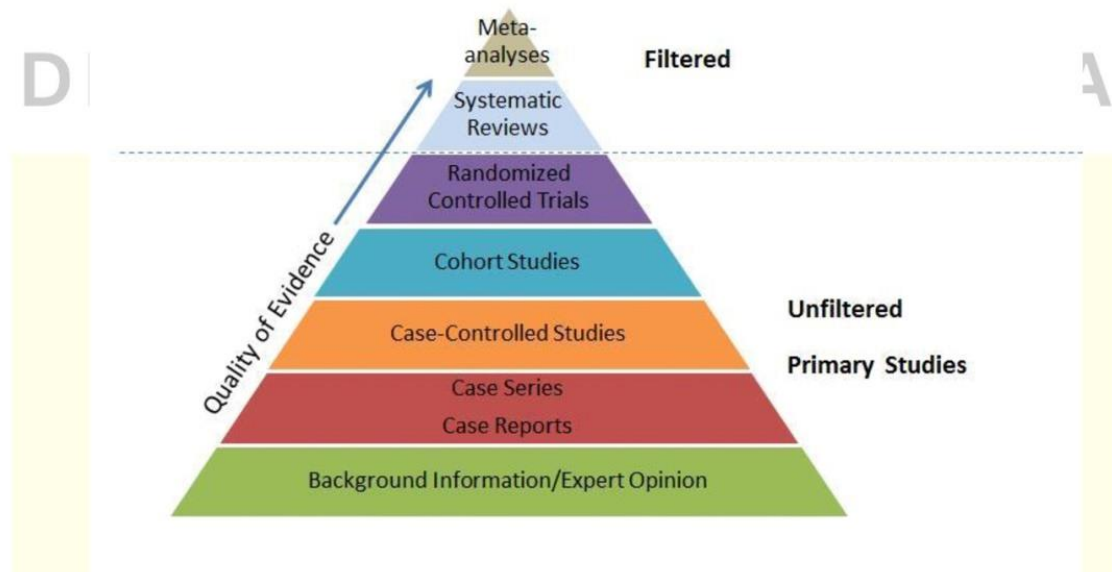
✓ **RR <1** → Exposure **NEGATIVELY** related to disease(outcome)

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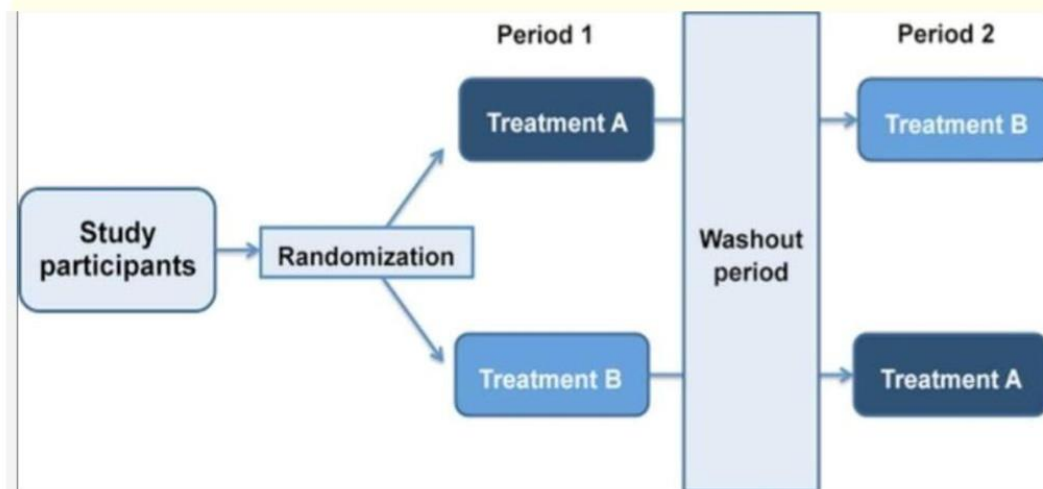
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➤ The **RCT (randomized control trial)** is the **gold stander study** and the most evident one .

❖ **Types of RCT**

- A. parallel
- B. cross over



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⇒The Study design can be divided into two main types

1. **Descriptive study**
2. **Analytical study**, with their subtypes.

❖ **Descriptive study**

We are trying to **describe** the **outcome** of something among a specific **sample** or group of **people** (only describing what is happening) – like describing its number or other things by time ,place and person.

❑ Establish **the prevalence**.

❑ **No** control group.

* **(PO)** = (P : the population or the group sample) , (O : the outcome)

❑ **Example** : The prevalence of insomnia among medical student

P= Medical student , O = insomnia

*So we are trying just to **Describe how is common (prevalent) is insomnia among medical student**

* **We are just Observing the outcome !**

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❖ **Analytical study (PECO)**

🔍 We are trying to **Analyze** or **study** the **relationship** between an **Exposure** and an **Outcome**.

So, there must be **an exposure or intervention and an outcome** that we are looking to see if the study sample is exposing to it.

* **(PECO)** = (P : population or group sample), (E: Exposure or intervention), (C: comparison group), (O: outcome)

Example : The prevalence of insomnia among medical student and its effect on academic performance Or The effect of insomnia on academic performance among medical student

P = medical student, E= insomnia, C= group with no insomnia, O= academic performance

So here we are trying to **analyze the relationship between insomnia and academic performance** , is there an effect or no ? is it a harmful effect or protective effect ?

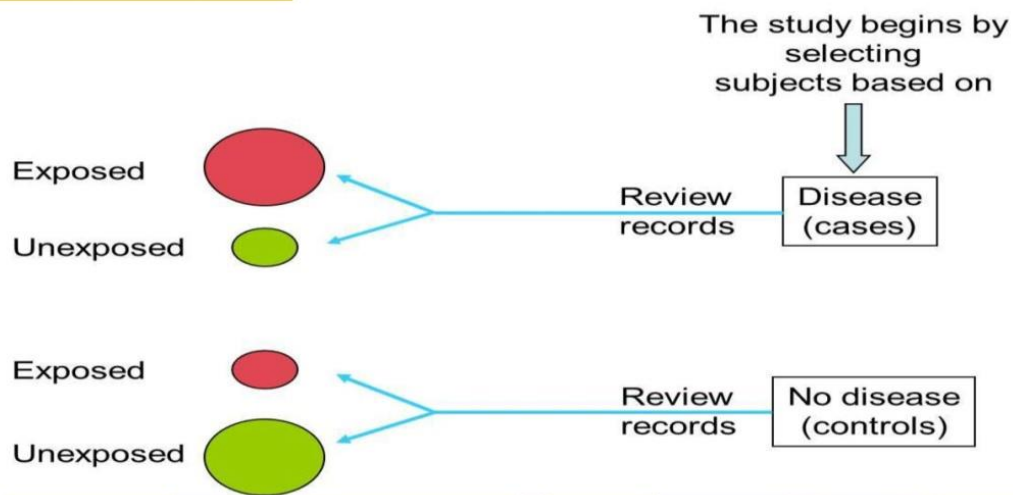
**** Note** : The exposure can be a drug , risk factor, education or other things that we can study its effect on something (outcome)

Case-Control Study

✓ **Based on the outcome** (Study begin from the outcome).



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There are **two groups** ,one with outcome and the other without outcome (control).

➤ The used measure in **Case-Control** study is **Odds ratio (OR)**
 OR = chance of event happening / chance of non event happening .

➤ Interpretation of the Odds Ratio:

- ✓ OR = 1 → Exposure **NOT** related to disease(outcome)
- ✓ OR >1 → Exposure **POSITIVELY** related to disease(outcome)
- ✓ OR <1 → Exposure **NEGATIVELY** related to disease(outcome)

* Note : as odds ratio increase , as the strength of relation between E+O is more .

Advantage	Disadvantage
Easy and quick	Cannot generate incident data
Cheap and has good rank	Bias (specially recall) is common
Good for rare disease	Control selection can be difficult

⇒ Cohort Study

Based on **exposure** (Study begin from the exposure).

✓ There are **two groups completely healthy or without outcome**, one is exposed and the other not exposed (control) , then follow the two groups in the future to see the outcome.

❖ Two designs:

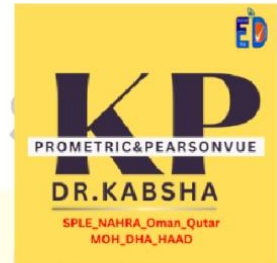
1. Prospective cohort design.
2. Retrospective cohort design.

➤ The used measure in Cohort study is **Relative Risk (RR)**.

RR = incidence of outcome in exposed / incidence of outcome in non exposed

Interpretation of the Relative Risk :

- ✓ RR = 1 → Exposure **NOT** related to disease(outcome)
- ✓ RR > 1 → Exposure **POSITIVELY** related to disease(outcome)
- ✓ RR < 1 → Exposure **NEGATIVELY** related to disease(outcome)



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Study Types

Study type	strengths	Limitations
A. Observational and descriptive Studies		
Case series, Case report Case reports and series are helpful in recognizing and describing new disease processes or rare manifestations and identify emerging health conditions. "documents unusual adverse events observed in single patients"	1. Help describe rare manifestations and new diseases. 2. Identify emerging health conditions. 3. Better suited to describing the characteristics of the disease than identify causality.	1. Purely descriptive and considered to be the weakest form of evidence. 2. Lack of control group (no comparison) 3. Can not be used to test for the presence of a valid statistical association. 4. Based on the experience of only one person.
Cross-sectional study Prevalence study "It is a <u>survey</u> of the total population for the exposure and disease (or drug use problems) at a <u>single point in time</u> ".	1. Fast and inexpensive. 2. Avoids the problem of loss to follow-up. 3. A cross-sectional study conducted at the beginning of a cohort or clinical trial provides demographic and clinical characteristics at baseline.	1. Cannot estimate incidence 2. Difficult to establish causal relationship 3. Impractical for the study of rare diseases and requires a large sample size 4. Only a snapshot at a time leading to a misinformation 5. The result <u>not</u> representative of the whole population (only individuals who participate in the study).
B. Observational and analytical Studies		
Case-control study Compare cases with a disease to controls without the disease, looking for differences in exposures (multiple possible causes of a single disease).	1. Relatively inexpensive because of smaller sample sizes. 2. It is useful for studying a relatively rare disease 3. Multiple etiologic factors can be studied. 4. Useful for generating hypotheses about the causes of an outcome variable. 5. One does not need an ethical consent. 6. There is no risk to the subject	1. Low internal validity (data collection is based on event recall). 2. High chance of bias because of separate sampling of cases and controls and retrospective measurement of various predictor variables. 3. <u>Cannot estimate incidence or prevalence.</u> 4. Only one outcome can be studied, whereas cohort and cross-sectional studies can evaluate multiple outcome variables
Cohort study Can be prospective or retrospective In a prospective cohort study, patients are identified at the beginning of the study based on their exposure status (exposed or not exposed) and followed up for a period from exposure to outcome	1. Can measure incidence. 2. Establish temporal association 3. Good for common diseases. 4. Good for rare exposures. 5. Possible to study associations of an exposure with several outcome variables.	1. Expensive because of large sample size and long follow-up. 2. Not good for rare diseases. 3. Confounding factors 4. A number of potential biases must be taken into account in conducting cohort studies. a. Bias in assessment of the outcome b. Information bias or recall bias. c. Biases from lack of response due to loss to follow-up. d. Analytic bias.
• Both designs are identical, comparing exposed and non-exposed populations. The only difference is calendar time. • The duration of follow-up and cost, associated with prospective cohort studies may be reduced by conducting retrospective cohort studies.		

C. Intervention Study (Experimental)		
Randomized controlled study Subjects in a population are randomly allocated into groups, usually called study and control groups to receive and not receive an experimental preventive or therapeutic procedure. "The investigator applies an intervention and observes the effect on one of more outcomes".	1. Exposure is under the control of investigator. 2. Randomization 3. Blinding eliminates bias 4. Control on time span 5. Confounding factors can be controlled 6. Best method to study causal relationship 7. We can confirm or refute etiological hypothesis on evidence. 8. More than one outcome can be measured.	1. Expensive 2. Subject exclusion may limit ability to generalize findings to other patients. 3. A long period of time is often required to reach a conclusion. 4. A large number of participants may be required. 5. Need an ethical consent. 6. Subjects may not comply with treatment assignments. 7. May expose participants to potential harm.

Questions

1- If a study is to follow a group of smokers and a group of nonsmokers time, and see which ones eventually develop lymphoma and which. What do you call this study design?

- A. Retrospective cohort study
- B. Prospective cohort study
- C. Cross-sectional study
- D. Case-series

2.....Is a technical managerial function that enables to deal with the present and anticipate the future?

- A. Planning
- B. Organizing
- C. Staffing
- D. Controlling

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3- Which one of the following designs could be deemed a study?

- A. Cohort
- B. Clinical trials
- C. Case-control
- D. Cross-sectional

4.....Is a technical function;it means establishing authority and responsibility relationships and formal structure and reporting relationships ?

- A. Planning
- B. Organizing
- C. Staffing
- D. Controlling

5- Which of the following is an interventional study?

- A. Cohort
- B. Ecological
- C. Case-Control
- D. Randomized Clinical Trial

6.....Is a technical function on monitoring, adjusting and improving performance?

- A. Planning
- B. Organizing
- C. Staffing
- D. Controlling

7- Researchers compare a group of people who have diabetes with of people who have no diabetes, and looks back in time to see characteristics of the two groups differ.Which of the following is the described study design?

- A. Prospective cohort
- B. Case-Control
- C. Experimental
- D. Randomized clinical



8....Is the acquiring and training human resources?

- A. Planning
- B. Organizing
- C. Staffing
- D. Controlling

9- Which of the following studies are more susceptible to recall bias?

- A. Retrospective cohort
- B. Prospective cohort study
- C. Randomized controlled trial
- D. Non-randomized controlled trial

10- Which of the following is the type of studies considers data from multiple studies of different designs to draw conclusions?

- A. Case Series
- B. Double-blind study
- C. Systematic Review
- D. Consensus statement

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11- The authors report two isolated gastro-intestinal tuberculosis in renal transplant recipients that illustrates the difficulty of making this diagnosis and a brief review of the literature on its clinical presentation, diagnosis and therapeutic approach.

What is the type of study in the above-mentioned abstract?

- A. Case-series
- B. Cohort studies
- C. Case-control studies
- D. Cross-sectional studies



12....Is a social behavior in nature and focus on initiating action in the organization ?

- A. Planning
- B. Organizing
- C. Directing
- D. Controlling

13- Scenario: A study located a group of subjects with lymphoma and went to identify which subjects are smokers and which are not.

Which of the following study designs is used to conduct the study in the scenario?

- A. Case-series
- B. Cross-sectional study
- C. Prospective cohort study
- D. Retrospective cohort study

14.....Is a technical management function that is a part of all other management function?

- A. Planning
- B. Organizing
- C. Decision making
- D. Controlling

15- Which of the following terms measures the association between the exposure and the outcome?

- A. P Value
- B. Risk ratio
- C. Standard deviate



16- Which of the following study design is the best to use to assess the awareness on the usage of vitamin D supplements among people of Jeddah

- A. Cohort study
- B. Case report study
- C. Case control study
- D. Cross sectional stud

17.....Is a process , composed of interrelated social and technical function and activities (include roles) ,occurring in a formal organizational setting for the purpose of accomplishing predetermined objectives through the utilizing of human and other resources?

- A. Management
- B. Leadership
- C. Directing
- D. Decision

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18- A researcher gathered all vitamin E studies from the past ten years. Vitamin E was used for a variety of condition. The populations studied as well as the vitamin E formulations and doses were all different. The researcher compared the incidences of cardiovascular-related mortality in those taking vitamin E supplements versus those that did not.

Which of the following best describe this type of study ?

- A. Meta-analysis
- B. Cohort study
- C. Observational study
- D. Controlled clinical trial



19- A group of patients with diabetes were interviewed to determine they are currently eating fast food or not. Which epidemiological study design is this?

- A. Cohort
- B. Case-control
- C. Cross-sectional
- D. Randomized control trial

20....Is the ability to engage in a set of behaviors that are functionally related to one another and that lead to a desired performance level in a given area ?

- A. Management
- B. Leadership
- C. Directing
- D. Skill

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21- Which of the following is the disadvantage of conducting randomized Controlled trial ?

- A. Expensive
- B. High possibility of bias
- C. Difficult to control confounders
- D. Cannot be used for hypothesis testing

22- Study that investigate if drinking coffee lead to coronary heart found that there is a correlation between two variables. Study that investigate if drinking coffee lead to coronary heart found that there is a correlation between two variables. However drinkers are more often smokers than the average and, correlation between drinking coffee and nicotine consumption, strong causal correlation between smoking and the incidence of heart disease. Which of the following type of bias affect this study results?

- A. Measurement errors
- B. Hawthorne effect
- C. Information bias
- D. Confounding

Co-founding

The factor that's associated with exposure ...and cause false relation between exposure & outcome.

23....are related to the ability to visualize the organization as a whole ,recognize relationships as among organizational parts, and understand how the organization fits into the wider context of the industry ,community ,and the world ?

- A. Human relation skills
- B. Conceptual skills
- C. Technical skills
- D. Clinical skills

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26-researcher was interested in examining the association between postmenopausal hormone Replacement therapy (HRT) and development of heart disease. All women who were characterized a postmenopausal were approached regarding their interest in participating in the study by answering a questionnaire annually regarding their medication use and medical conditions. Of the 16,168 women who provided consent, the average length of follow-up was 12.5 years (range, 6 to 16 years). Which of the following best describes the study design?

- A. Case-control study
- B. Prospective cohort study
- C. Randomized controlled trial
- D. meta analysis

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Child birth weight variable is used as ?

- A) Ordinal
- B) Nominal
- C) Binary
- D) Continuous



(Yes or no)data type is?

- A)nominal
- B)ordinal
- C)interval
- D)ratio

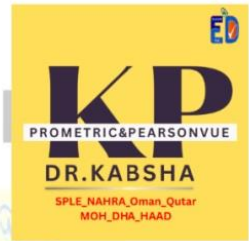
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❖ **Bias**

The process that leads to deviation of the result from the truth



Random error

- Randomly in all direction
- Inevitable to any study
- Can be reduced by increasing the sample size



Systematic error or Bias

- Systematically in a particular direction
- When there is Methodological flaw
- Cannot be reduced by increasing the sample size



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Selection bias? Random selection

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❖ Bias

Definition

Systematic, non-random variation in study methodology and conductance, introducing **error in outcome** interpretation.



Bias can occur in all aspects of the study design.

Examples:

1. **Selection bias:** (sampling bias or enrolling bias)

- an error in **choosing** the individuals or groups to take part in a scientific study
- misuse of inclusion and exclusion criteria

E.g. selecting geriatrics participant to test the relation between aminoglycosides & renal failure

➡ **To overcome**----- (clear and sharp inclusion and exclusion criteria)

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2. Observation or information bias

Occurs during studies in which the **researcher expects a certain answer.**

e.g. if during a medical study the doctor knows whether the volunteer is part of the control, he or she will look for certain symptoms.

When someone expects a certain outcome they are more likely to find evidence that supports that expectation than look carefully for contradictions.

➔ **To overcome**----- **blinding**

N.B.----- **open** label & **single** blinded & **double** blinded & **triple** blinded

3. Recall bias

- 1- Rely on memory of participants
- 2- Usually in retrospective studies
- 3- E.g., studies of birth defects secondary to medications

➔ **Overcome** ---- By depending on **documents** not the memory

4. interviewer Bias

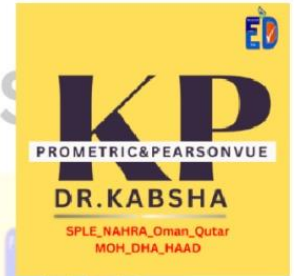
e.g., interviews are not conducted in a uniform manner for all study participants.

➔ **Overcome**--- By patient self-assessment

5. attrition bias --- loss to follow up

6. misclassification bias

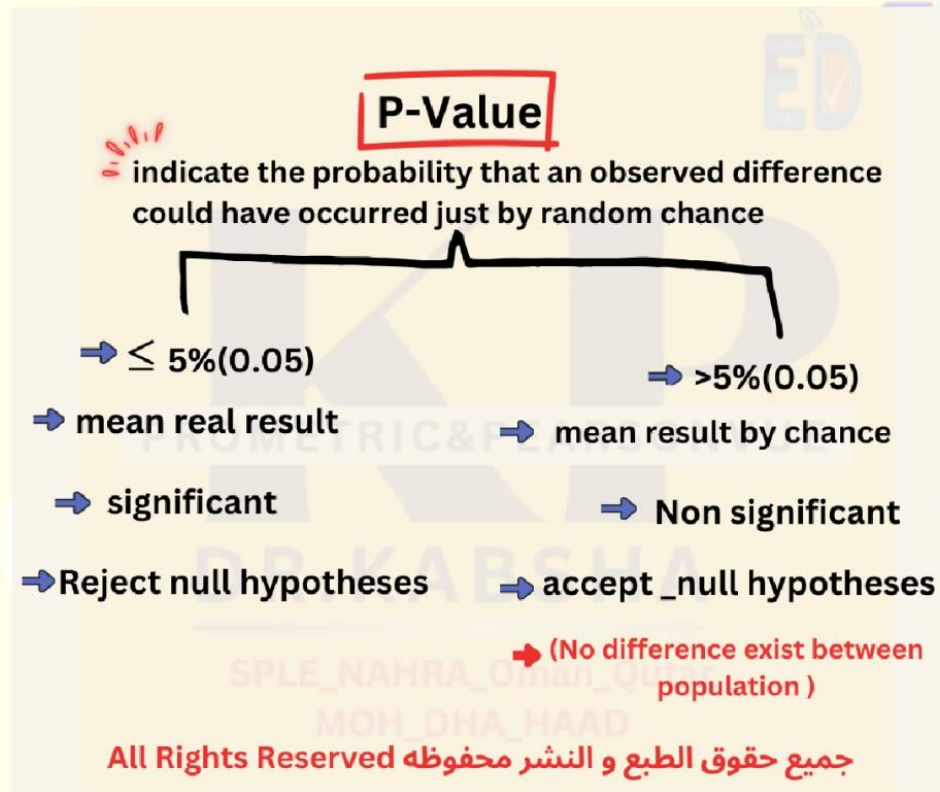
It occurs when an individual is assigned to a different category than the one to which they should be assigned (1). For example, if a patient appears to be non-hypertensive because of modification-controlled blood pressure, resulting in systolic and diastolic measures that are within the normal range, this may constitute an incorrect classification.



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7. **Reporting bias**

If researchers find their hypothesis was wrong they don't publish it



- H0 null hypothesisno difference exist between population
- H1 alternative hypothesis Difference exist between population

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❖ **Definition of Validity**

The extent to which a concept, conclusion, or measurement can correspond accurately to the real world. Also, to what extent the study measures what it is set to do. (e.g., were the variables controlled, are the samples representative).

Two types of validity:

- **Internal validity**

To extent to which the observed difference between groups can be attributed to the intervention being investigated.

- **External validity:**

Applicability of the study findings to the real-world patients.

❖ **Definition of Credibility**

A measure of the truth value of qualitative research, or whether the study findings are correct and accurate. (e.g., is it a trustworthy source, what is the expertise of the author).

❖ **Definition of Reliability**

Consistency of the results (if we repeat experiment would we get same results)

Two types of reliability:

- **Primary**

If the same study was repeated in a similar sample would we get the same results?

- **Secondary**

The findings consistent with other published studies investigating the same thing

❖ **Definition of NNT**

The Number Needed to Treat (NNT) is the number of patients you need to treat to prevent one additional bad outcome (death, stroke, etc.). For example, if a drug has an NNT of 5, it means you have to treat 5 people with the drug to prevent one additional bad outcome

NNT number need for treatment if drug

A = 40

Drug B = 20

Drug c = 10

Drug D = 9

The most potent drug?

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Median (Middle) The number which is in the middle or the middle value. 11 7 11 18 9 7 6 23 7 6 7 7 7 9 11 11 18 23 Median: 9	Mode (Most) The number that appears the most. 11 7 11 18 9 7 6 23 7 6 7 7 7 9 11 11 18 23 Mode: 7
Mean (Average) The total of the numbers divided by how many numbers there are. 11 7 11 18 9 7 6 23 7 11+7+11+18+9+7+6+23+7=99 99 / 9 = 11 Mean: 11	Range (Difference) The difference between the largest and the smallest number. 11 7 11 18 9 7 6 23 7 Large : 23 Small : 6 23 - 6 = 17 Range: 17

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$$\text{Mean} = \frac{\text{Sum of Observations}}{\text{Total Number of Observations}}$$

If 'n' is odd: $\text{Median} = \left(\frac{n+1}{2} \right)^{\text{th}} \text{ term}$

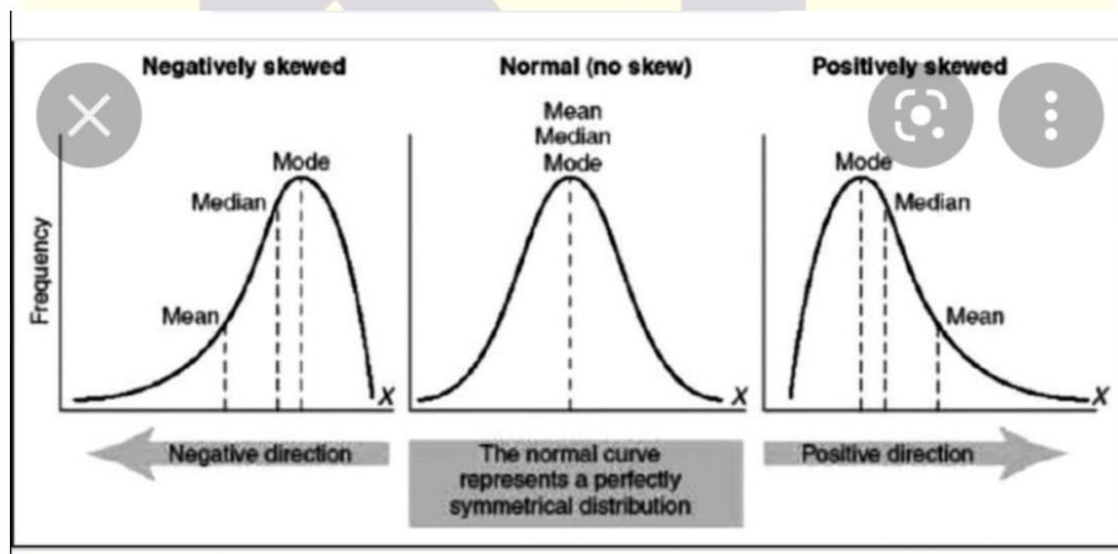
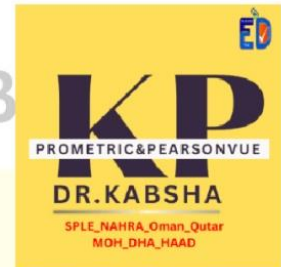
If 'n' is even: $\text{Median} = \frac{\left(\frac{n}{2} \right)^{\text{th}} \text{ term} + \left(\frac{n}{2} + 1 \right)^{\text{th}} \text{ term}}{2}$

2) From the following data of the highest in cm of 11 pediatric patients:

103, 107, 110, 124, 129, 131, 136, 140, 142, 145, 150

what does the value (131) represent?

- A. Mode
- B. Mean
- C. Median
- D. Confidence interval



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